# DEGRACE companions in MG-GLAM: status

**What landed, what the products are, how they were verified, what the pilots say, what is left**

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17 October 2026 merged into `cz`; plan in `docs/`

# One week from plan to merged code

- **Scope** (plan of 16 Oct): 42 exact outputs per box (17 primary redshifts + 25 stations at $\Delta\chi = 256\,h^{-1}$Mpc), lens planes instead of snapshots, and the extra products of the brief: $\delta_L$, low-$k$ density and momentum modes, a hashed subsample, $f_R$ at halo centres.
- **Implemented** on `degrace-track-a` (17 commits, 53 files, +10.8k lines): items I1–I10; I11 (optional) not done.
- **Verified**: 180-check regression suite + four unit/integration scripts, six independent code reviews (0 blockers, 8 majors, all fixed), two pilots on ECOSMOG node 2.
- **Merged** into `cz` today (fast-forward, suite green on `cz`).

**Take-away**

Everything a companion run needs exists, is tested, is on the mainline and has run twice at $1024^3$. Three design decisions and one $0.5\,\%$ question remain.

# Status of the plan items

| item | what | status | evidence |
|---|---|---|---|
| I1 | exact-$z$ scheduler accepts arbitrarily close output pairs | **done** (`cz` ae4fa9d) | suite case 7 |
| I2 | lens planes: 12 per output, $\Sigma/\bar{\Sigma}$ on $2048^2$ | **done** | L1–L3 |
| I3–I5 | input generator, `OutputSchedule.dat`, per-output classes | **done** | cases 9, 4e; S2 (42 epochs = ECOSMOG to $10^{-7}$) |
| I6 | provenance per box | partial | JSON lacks generator/LPT/seed (minor) |
| I7 | $\delta_L(k)$ dump, route 1: in our FML IC generator | **done** | V3, V7, V8; `run_deltaL_test.sh` |
| I8 | low-$k$ $\delta_m$ and $(1+\delta)$v modes | **done** | V1, V2, mode-by-mode |
| I9 | ID-hashed quantised subsample, 1 in $M$ | **done** | V5, V6; `run_ic2pm_test.sh` |
| I10 | $f_R$ at BDM halo centres | **done** | unit test + f(R) smoke case |
| I11 | optional extras | **open** | not started |
| pilot | node 2 $\Lambda$CDM, $N_g = 2048$ | **done** | verified, 0 FAIL |
| pilot | node 2 f(R), extras + subsample + $f_R$ | **done** | verified, 0 FAIL; 6 h 08 min |

All flags default off: without the new `Init.dat` lines the code follows the `cz` path (review 2), apart from `OutputSchedule.dat` and an appended `Setup.dat` block.

# Schedule and output classes (I1, I3–I5)

- **17 primary redshifts** $z = 0, 0.125, \ldots, 2$ ($\Delta z = 0.125$), common to all cosmologies.
- **25 stations** at comoving distance $\chi = 256\,k\,h^{-1}$Mpc, $k = 1 \ldots 25$, from the run's own flat matter + $\Lambda$ $\chi(a)$ (no radiation, $c/H_0 = 2997.92458\,h^{-1}$Mpc); `gen_inputs.py` reproduces every ECOSMOG list to $10^{-7}$.
- `#outputs = -42`: every epoch is hit exactly (a step moved onto it or inserted); pairs closer than 0.5 % are allowed. Pilot: 181 steps = 163 ladder + 18 inserted, $\mathrm{d}a_0 = 6 \times 10^{-4}$ from $z = 49$.
- `OutputSchedule.dat`: one line per output: step (= snapshot number), $a$, $z$, $\chi$, exact, class.
- **Class 1** (full): $P(k)$, planes, BDM catalogue, $f_R$ at haloes, $\delta_m$ and momentum cubes, subsample; **class 2** (station): $P(k)$ and planes. Pilots 17 + 25; production: all 42 class 1.

| step | $a$ | $z$ | $\chi$ | exact | class |
|---|---|---|---|---|---|
| 57 | 0.10275 | 8.732 | 6400.0 | 1 | 2 |
| 63 | 0.11832 | 7.452 | 6144.0 | 1 | 2 |
| 105 | 0.33333 | 2.000 | 3625.9 | 1 | 1 |
| 106 | 0.33796 | 1.959 | 3584.0 | 1 | 2 |
| 127 | 0.50000 | 1.000 | 2312.7 | 1 | 1 |
| 128 | 0.50128 | 0.995 | 2304.0 | 1 | 2 |
| 152 | 0.72727 | 0.375 | 1024.9 | 1 | 1 |
| 153 | 0.72747 | 0.375 | 1024.0 | 1 | 2 |
| 181 | 1.00000 | 0.000 | 0.0 | 1 | 1 |

Node 2 ($\Omega_m = 0.3$), $\chi$ in $h^{-1}$Mpc. Epochs equal the pinned ECOSMOG list to $10^{-7}$; steps 105/106, 127/128, 152/153 are the close pairs.

# What a box writes: files per output

| file (<step> = snapshot, box 0001) | content | $1024^3$, $N_g = 2048$ (pilots) | $2048^3$, $N_g = 4096$ |
|---|---|---|---|
| `PowerDM.R/log.<step>.0001.dat` | $P(k)$ on the $N_g$ mesh, linear / log bins, raw CIC | 41 KB | 80 KB |
| `LensPlanes.<step>.0001.bin/.meta` | 12 planes of $\Sigma/\bar{\Sigma}$, $N^2_{\rm lens}$ float32 | 201 MB ($2048^2$) | 805 MB ($4096^2$) |
| `CatshortV.<step>.0001.DAT` / `.bin` + `.meta` | BDM haloes (in `CATALOGS/`): text, 24 columns; binary twin, 96 B per halo (`CatalogueFormat`) | 0.1–0.3 GB (binary $\frac{1}{3}$) | 0.3–0.4 GB (binary $\frac{1}{3}$) |
| `deltam.<step>.0001.bin/.meta` | $\tilde{\delta}_m(\mathrm{k})$ cube, $k_c = 1.5$ | 469 MB | 469 MB |
| `mom_x,y,z.<step>.0001.bin/.meta` | $[(1+\delta)v_i]^\sim(\mathrm{k})$ cubes, $k_c = 0.75$ | $3 \times 59$ MB ($3 \times 138$ at $k_c = 1$) | same |
| `Subsample.<step>.0001.bin/.meta` | 1 in $M = 100$ particles, int16 | 129 MB | 1.03 GB |
| `fR_halo.<step>.0001.bin/.meta` | $(f_R^{\rm centre}, f_R^{\rm mean})$ per halo | 9 MB (1.1 M haloes) | ∼30 MB |
| `<IC prefix>_deltaL.bin/.meta` | $\tilde{\delta}_L(\mathrm{k})$ cube, once per box, from the IC job | 469 MB | 469 MB |
| `OutputSchedule.dat`, `timing.log` | schedule; per-step timing (with `Run.log`) | KB | KB |

- No particle snapshots (`iSave = 0`). Per box: 9.1 GB (ΛCDM pilot, no extras), 25 GB (f(R) pilot, extras at the 17 primaries); with the extras at all 42 outputs $\approx 40$ GB at $1024^3$ and $\approx 130$ GB at $2048^3$ (13 TB for 100 boxes).

# Conventions shared by every product

- **Epochs.** Positions at the output $a$ ($a_x = a$); velocities half a step behind, $a_v = a - \Delta a/2$ with $\Delta a$ the step just taken (leapfrog). Every sidecar records `step, a, z, a_x, a_v`. The legacy BDM catalogue converts its velocities with $1/a$ instead of $1/a_v$ (0.5–1.1 % high; decision pending).
- **Units.** Comoving $h^{-1}$Mpc, peculiar velocities in km/s, masses in $h^{-1}M_\odot$; axes follow the particle $x, y, z$ of the catalogue (`axis_map`).
- **Fourier.** $\tilde{\delta}(\mathrm{k}) = N^{-3}\sum_{\mathrm{x}} \delta(\mathrm{x})\, e^{-i\mathrm{k}\cdot\mathrm{x}}$, $\mathrm{k} = (2\pi/L)(i_x, i_y, i_z)$, $P(k) = L^3\langle|\tilde{\delta}|^2\rangle$; grid node $i$ at $(i-1)L/N$.
- **Mode cube.** $|i_x|, |i_y| \le n$, $0 \le i_z \le n$ with $n = \lfloor k_c L/2\pi \rfloor$; the Hermitian-redundant $i_z = 0$ modes are kept, $\mathrm{k} = 0$ is zero; $(2n+1)^2(n+1)$ complex64 values.
- **Layout.** Fortran `C(0:n,-n:n,-n:n)`, $i_z$ fastest; numpy `fromfile(f, '<c8').reshape(2n+1, 2n+1, n+1)[ix+n, iy+n, iz]`.
- **Mass assignment.** CIC on $N_{\mathrm{dep}}$ nodes, never deconvolved inside GLAM; `assignment`, `N_dep` and `interlaced = false` are recorded and offline code divides by $W_{\mathrm{CIC}}$.
- **Sidecar keys.** `product, format, layout, fourier_norm, modes, n, k_c, N_dep, N_grid, N_row, N_p, Box, units, D_code(a)` (the code's flat matter + $\Lambda$ growth, $D(1) = 1$), `Omega_m, Omega_L, hubble, MG_flag, MG_model, fR0`, and `header` (the PM header: `ic2pm` velocity epoch, lattice tag `IDx/IDz`).

# Lens planes (l2): definition, cost, checks

- **Definition.** For projection axis $X$ and slab $s$ of depth $L/N_{\rm slab} = 256\,h^{-1}$Mpc, pixel $(i, j)$ is the mean of $1 + \delta$ over the slab column and the pixel block, i.e. $\Sigma_s(i,j)/\bar{\Sigma}$ with $\bar{\Sigma} = \bar{\rho}_m \times$ depth; $\delta$ from the CIC deposit on $N_g$, not deconvolved. Each plane averages to 1 by construction.
- **Layout.** 12 planes in the order $X_1 \ldots X_4, Y_1 \ldots Y_4, Z_1 \ldots Z_4$; $X$-planes indexed $(y, z)$, $Y\,(x, z)$, $Z\,(x, y)$; pixel $(i, j)$ covers $[(i-1), i)\,L/N_{\rm lens}$; float32, column-major; block average when $N_{\rm lens} < N_g$.
- **Offline use.** $\kappa = \frac{3}{2}\Omega_m (H_0/c)^2 \sum_{\rm planes} \frac{\chi(\chi_s - \chi)}{\chi_s\, a}\,(\Sigma/\bar{\Sigma} - 1)\,\Delta\chi$, $\Delta\chi = 256\,h^{-1}$Mpc.
- **Cost.** 1.8 s per output, the 201 MB write 12 s: $< 1\,\%$ of a run.

| check | result |
|---|---|
| L1 mass conservation per axis | $10^{-7}$ |
| L2 vs `ProjectDENSIT` | $10^{-5}$ |
| L2 $36^2$ = block mean of $72^2$ | $10^{-5}$ |
| L3 $P_{2D}$ vs projected $P_{3D}$ | $3\sigma$ per ring; 3–7 % |
| pilots: plane means | $1 \pm 10^{-7}$ |
| pilots: $\Sigma/\bar{\Sigma}$ range at $z = 0$ | 0.10–51 |

**Open question for Baojiu**

Plane normalisation ($\Sigma/\bar{\Sigma}$ vs $\Sigma$); do the stations also need halo catalogues?

# Fourier dumps, subsample, $f_R$ at haloes (l8–l10)

| product | what | verified |
|---|---|---|
| `deltam` | $\delta_k = N^{-3}\sum_x \delta(x)e^{-ik\cdot x}$, cube $\lvert k_i\rvert \le k_c = 1.5$, CIC on $N_g$, complex64 $C(0{:}n, -n{:}n, -n{:}n)$ | V1 layout; V2 = `PowerDM.R` to $10^{-4}$ |
| `mom_x,y,z` | $(1+\delta)\,v$ on $N_{\rm dep} = N_g/2$, $v$ at $a_v = a - \mathrm{d}a/2$, $k_c = 0.75$ (below $k_{N,\rm dep}/4$: no interlacing) | V6; mode by mode vs decoded particles $7 \times 10^{-5}$ |
| `Subsample` | keep 1 in $M$ by splitmix64(ID $\oplus$ salt), int16 positions and velocities, deterministic across twins, threads, restarts | V5 ($10^7$ IDs), V6 |
| `fR_halo` | $f_R = f_{R,\rm bg}(a_{\rm old})\,u^{n+1}$ at every BDM centre | unit test; f(R) smoke case |

- Storage at $1024^3$ per output: $\delta_m$ 0.47 GB, momentum 0.18 GB ($k_c = 0.75$; 0.41 at 1.0), subsample ($M = 100$) 0.13 GB.
- From the review: `ic2pm` places every particle at array index = Gadget ID and tags the lattice order (`IDz` 2LPTic, `IDx` FML) in the PM header, so the subsample index gives the Lagrangian position.

# Definitions: density, momentum and $\delta_L$ cubes (I7, I8)

- `deltam`: $\delta(\mathrm{x}) = n(\mathrm{x})/\bar{n} - 1$ from the CIC deposit of the output positions on $N_{\rm dep} = N_g$, forward FFT, cube $|k_i| \le k_c = 1.5\,h\,\mathrm{Mpc}^{-1}$ ($n = 244$ at $L = 1024$: 0.47 GB). Offline $P_{mm}(k) = L^3\langle|\tilde{\delta}_m/W_{\rm CIC}|^2\rangle$ equals `PowerDM.R` bin by bin to $10^{-4}$ (V2).
- `mom_x,y,z`: $p_i(\mathrm{x}) = \bar{n}_{\rm cell}^{-1}\sum_p v_{i,p}\,W_{\rm CIC}(\mathrm{x} - \mathrm{x}_p) = (1+\delta)\,v_i$, $v$ the peculiar velocity in km/s at $a_v$; deposit on $N_{\rm dep} = N_g/2$, $k_c = 0.75$ by the rule $k_c \le k_{\rm N,dep}/4 = \pi N_{\rm dep}/4L$ (no interlacing; the f(R) pilot used 1.0). At linear $k$, $\tilde{v}_i \simeq \tilde{p}_i$.
- `deltaL`: the IC generator's Gaussian field at $z_{\rm ini}$ divided by $D(a_{\rm ini})$ (flat matter + $\Lambda$, $D(1) = 1$), i.e. the linear field at $a = 1$; written before any LPT displacement, same cube and layout as `deltam`, identical for a $\Lambda$CDM/f(R) twin pair (one IC file). Sidecar adds `seed, z_ini, D_gen(a_ini), pofk_file, lpt_order, fix_amplitude, lattice` (particles at $iL/N_p$, ID $= 1 + i_x + N_p(i_y + N_p i_z)$) and the cube rms.
- **Uses.** Halo propagator $\Gamma_h(k|M) = \sum \mathrm{Re}[\tilde{\delta}_h\tilde{\delta}_L^*]/\sum|\tilde{\delta}_L|^2$ without cosmic variance; matter propagator $G_m = P_{mL}/P_L$; growth check $\langle\tilde{\delta}_m/W_{\rm CIC}/\tilde{\delta}_L\rangle = D_{\rm code}(a)$ (0.9997–1.0012 at $k \le 0.15$, V3); halo–momentum cross spectra; environment and tidal fields at $R \ge 3/k_c$.
- **Caveat.** Compare dumps with binned means and cross-correlations, not single-mode ratios (CIC of lattice particles is not linear in the displacement; phase scatter $\sim k\Delta/2$ at early times).

# Definitions: particle subsample (I9) and $f_R$ at haloes (I10)

`Subsample`

- ID = 0-based lattice index $i + N(j + Nk)$ ($i$ along $x$ for FML ICs, tag `IDx`; Baojiu's 2LPTic files are $k$-fastest, `IDz`); `ic2pm` places particle = Gadget ID. Lagrangian position $\mathrm{q} = (i, j, k)\, L/N$.
- Keep iff `splitmix64(ID xor SALT) mod M = 0`, `SALT = 0x5EED00DE6C0A2026`, $M = 100$: the same particles at every output and in both twins; $n_{\rm sel} = 10\,741\,150 = N/100$ to 0.03 %.
- Stored in increasing ID: the $k$-th particle is the $k$-th ID passing the rule (IDs implicit).
- int16 quantisation: $q = \lfloor 65536\, x/L \rfloor - 32768$, decode $x = (q + 32768 + 0.5)\, L/65536$ ($\Delta x = 15.6\, h^{-1}$kpc); $s = \mathrm{nint}(v/\Delta v)$, $\Delta v = 0.25$ km/s, clipped at $\pm 32767$ (count in the sidecar, 0 in the pilot); $v$ peculiar at $a_v$.
- Layout: `qx, qy, qz, vx, vy, vz`, $n_{\rm sel}$ each, contiguous; 12 B per particle.

`fR_halo` (f(R) runs, `MG_model = 1`)

- One float32 pair $(f_R^{\rm centre}, f_R^{\rm mean})$ per catalogue row, in `CatshortV` order.
- $f_R^{\rm centre}$: CIC interpolation of $f_R$ at the BDM centre; $f_R^{\rm mean}$: mean over the mesh cells whose centres lie within $R_{\rm vir}$ (equal to the centre value below half a cell).
- $f_R(\mathrm{x}) = f_{R,\rm bg}(a_{\rm old})\, u(\mathrm{x})^{n+1}$, $u$ the solver field (scalar field over its background value) at the pre-step epoch $a_{\rm old}$; $f_{R,\rm bg}(a) = f_{R0}\,[R_0/R(a)]^{n+1}$, $R(a) = 3(\Omega_m a^{-3} + 4\Omega_\Lambda)$.
- Sidecar: $f_{R0}$, $n$, $a_{\rm old}$, $f_{R,\rm bg}(a_{\rm old})$, catalogue name and row count.

# $\delta_L$ from the IC generator (I7, route 1)

- The FML generator writes `<prefix>_deltaL.bin/.meta`: its Gaussian field at $z_{\rm ini}$ divided by $D(a_{\rm ini})$, same layout and convention as `deltam`; $k_c = 1.5$, 0.47 GB.
- Lua: `write_deltaL`, `deltaL_kc`; on in the production input. Costs 1 s.
- Baojiu's ICs and the legacy pilots are out of scope (decision of 17 Oct).

| check | result |
|---|---|
| Zel'dovich: $\delta_L = -ik\cdot\Psi(k)/D$ | every mode to $10^{-3}$ |
| 2LPT ICs | residual $8\times10^{-3}$ (2nd order) |
| wrong axis / sign / half cell | off by $10^4\times$ |
| V3: ratio to $D(a)$, $k \le 0.15$ | 0.9997–1.0012 |
| V3: cross-correlation, $k \le 0.05$ | 0.99998 |
| V3: coordinate offset per axis | $\le 0.03\,h^{-1}$Mpc |
| V7: 1/2/4 MPI ranks | identical md5 |

**Lesson**

CIC density of particles sitting on grid nodes is not linear in the displacement (phase scatter $\sim k\Delta/2$ per mode at early times): compare dumps with binned means and cross-correlations.

# Definitions: BDM halo catalogue and $P(k)$ (legacy products)

- `CATALOGS/CatshortV.<step>.0001.DAT`: text, 8 header lines (PM header with the `ic2pm` epoch and lattice tag, $a$ and $\Delta a$, step, $N_{\rm row}$, $N_g$, cosmology, buffer width, mass of the smallest particle, overdensity limit), then one row per halo with 24 columns: $x, y, z$ [$h^{-1}$Mpc]; $V_x, V_y, V_z$ [km/s]; $M_{\rm bound}$, $M_{\rm tot}$ [$h^{-1}M_\odot$]; $R_{\rm vir}$ [$h^{-1}$kpc]; $V_{\rm rms}$, $V_{\rm circ}$ [km/s]; row index; $c_{\rm vir}$; $N_{\rm part}$; distinct/sub flag; $X_{\rm off}$; $2K/E_p - 1$; spin $\lambda$; RadRMS; $b/a$, $c/a$; major-axis direction $(x, y, z)$.
- **Binary twin** (`CatalogueFormat = text|binary|both` in `BDM.config`, default both; 2026-10-17): `CatshortV.<step>.0001.bin`, 96-byte little-endian records holding the same 24 columns (22 float32 + 2 int32), same rows and order, lossless and $3\times$ smaller; sidecar `.meta` with columns, units, rows, $a$, $a_v$, $\Delta a$, cosmology, $m_p$, `MassMin`, $\Delta_{\rm vir}$. Reader: `python/glamio` (`read_catalogue`).
- **Halo definition (BDM).** Centre = density peak refined by centre-of-mass iterations; radius = the Bryan–Norman virial aperture relative to the mean density ($\Delta_{\rm vir} = 338$ at $z = 0$, 203 at $z = 1$); `MassMin` $= 10^{11}\, h^{-1}M_\odot$ (particle mass $8.3 \times 10^{10}$ at $1024^3$); rows in peak order, not by mass; velocities converted with $1/a$ (the $a/a_v$ offset of the decisions frame).
- `PowerDM.R.<step>.0001.dat`: 2 header lines ($a$, step, $z$, $N_g$, $k_{\rm Ny} = \pi N_g/L$), then bin, $k$ [$h\,{\rm Mpc}^{-1}$], number of modes, $P$ [$(h^{-1}{\rm Mpc})^3$] in linear bins of width $2\pi/L$; $\delta$ from CIC on $N_g$, raw: divide by the shell-averaged $W^2_{\rm CIC} \simeq 1 - \frac{2}{3}\sin^2(\pi k/2k_{\rm Ny})$ offline (as in the twin comparison). `PowerDM.log`: the same in logarithmic bins.
- `timing.log`: per-step wall time split into Force, Move, Density, IO, BDM, Analysis, MG (source of the cost figures).

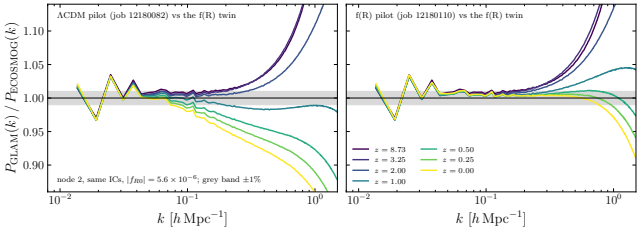
# Pilot: node 2 $\Lambda$CDM, $N_g = 2048$ (job 12180082)

| | |
|---|---|
| wall time | 2 h 32 min, 64 cores, `cosma8-serial` |
| CPU efficiency | 83 % |
| peak memory | 118 GB (91 GB `PMP2main`) |
| steps | 181, median 0.60 min |
| outputs | 42 $P(k)$, 42 plane sets, 17 catalogues |
| disk per box | 9.1 GB |
| core-hours | $\approx 165$ per $\Lambda$CDM box |

| step type | mean wall time |
|---|---|
| ordinary step | 0.60 min |
| station output: $2048^3$ $P(k)$ 25 s, planes 1.8 s, write 12 s | 0.96 min |
| primary output: + BDM halo finder | 2.6 min |
| all 42 outputs | 43 min (29 %) |
| of which lens planes | $< 1\,\%$ |

- `check_run.py`: schedule, epochs, files, sizes, plane means, catalogues, ic2pm epoch: 0 FAIL.
- BDM dominates the output cost; the new products are cheap. The extras (modes, subsample, $f_R$) add at most 1.4 min per primary output on the f(R) pilot (two frames on).

# Pilots vs the ECOSMOG twin: same ICs, mode-exact




- Left, ΛCDM vs the f(R) twin: flat $+0.5$ to $+0.7\,\%$ at $k < 0.1$ at every epoch, already at $z = 8.7$; late times: ECOSMOG above us by 5 % ($k = 0.1$–0.3) and 10 % ($k = 0.3$–1), the f(R) enhancement for $|f_{R0}| = 5.6 \times 10^{-6}$.
- Right, f(R) vs f(R), the real twin test: the same $+0.5$ to $+0.8\,\%$ at $k < 0.1$; within $\pm 1\,\%$ to $k = 0.3$ at every epoch; $-0.6\,\%$ at $k = 0.3$–1 at $z = 0$ (PM force resolution). $k > 0.3$ at $z > 3$: lattice discreteness.

## The $+0.5\,\%$ at linear scales is the same in both gravity models

A code or ingestion offset, not physics: `ic2pm`'s half-step convention, POWMES normalisation or a 0.25 % growth difference; a $\mathrm{d}a_0 = 3 \times 10^{-4}$ restart of 20 steps isolates the half-step term.

# f(R) pilot (job 12180110): completed and verified

| | f(R), extras on (ΛCDM pilot) |
|---|---|
| wall, 64 cores | 6 h 08 min (2 h 32 min) |
| CPU efficiency | 83 % (83 %) |
| MaxRSS | 177 GB (118 GB) |
| non-output step | 1.72 min; multigrid 1.1 min (0.60 min) |
| station output | $+0.37$ min: P(k), planes ($+0.36$) |
| primary output | $+2.9$ min: BDM + extras ($+2.0$, BDM) |
| 42 outputs | 59 min = 16 % of the run (29 %) |
| disk per box | 25 GB (9.1 GB) |

- `check_run.py`: 42 exact outputs, 42 P(k) + planes, 17 catalogues, 0 FAIL, 0 WARN.
- Per primary: $\delta_m$ cube $n = 244$ (469 MB), momentum cubes $n = 162$ ($3 \times 138$ MB), subsample of $10.74$ M particles $= N/100$ to 0.03 %, no clipping (129 MB), $f_R$ at 1.12 M haloes at $z = 1$ (9 MB): 1.0 GB.
- V9: the extras cost at most 1.4 min per primary (the excess over the ΛCDM primaries), $\leq 23$ min $= 6\,\%$ of this run and $\leq 15\,\%$ of a ΛCDM run; an upper bound until `ExtraOutputs` gets per-product timers. Memory 177 GB on a 1 TB node.
- $\approx 390$ core-hours per f(R) box at $1024^3$.
- Caveats: $k_{c,\mathrm{mom}} = 1.0$ (use the momentum modes below $0.78\,h\,\mathrm{Mpc}^{-1}$); the binaries predate the review fixes.

# Independent review: six reviewers, eight majors, all fixed

| review | B | M | m | n |
|---|---|---|---|---|
| 1 Fortran extras module | 0 | 0 | 9 | 5 |
| 2 scheduler, parameters, build | 0 | 2 | 6 | 6 |
| 3 particle subsample | 0 | 1 | 4 | 6 |
| 4 $\delta_L$ generator dump | 0 | 0 | 7 | 7 |
| 5 Python tools and tests | 0 | 2 | 10 | 6 |
| 6 docs and provenance | 0 | 3 | 11 | 8 |

Self-contained prompts and findings in `docs/reviews/degrace-track-a/`; rerunnable.

**Majors and their fixes**

- subsample "ID" was the IC file order $\rightarrow$ `ic2pm` scatters by Gadget ID and tags the lattice order
- optional `Init.dat` lines defaulted silently when malformed $\rightarrow$ stop with the line
- $f_R$ gate on `MG_flag` only $\rightarrow$ also `MG_model = 1`
- twin script skipped early epochs (5-decimal header) $\rightarrow$ epochs from the schedule file
- no test sensitive to axis order/sign of the dumps $\rightarrow$ mode-by-mode test
- plan flags table, catalogue velocity epoch, interlacing status $\rightarrow$ documented, rules added

**After the fixes**: suite 180/180, four scripts green, merged.

# Decisions pending and open items

**Decisions (yours)**

- **BDM catalogue velocities** are converted with $1/a$, the new products with $1/a_v$: a $1.1\,\%$ (first output) to $0.5\,\%$ ($z = 0$) offset. Fix the legacy finder (changes every GLAM catalogue) or keep documenting it (apply $a_v/a$ offline).
- **Momentum deposit** at $N_g = 4096$ f(R): the transient $N^3_{\rm dep}$ buffer (34 GB) does not fit the 985 GB runs; reuse `FI` or restrict extras to $N_g = 2048$.
- **Interlacing**: not implemented; default $k_{c,\rm mom} = 0.75$ stays below $k_{N,\rm dep}/4$ and PMP2init refuses more. Implement if $k > 0.78$ momentum modes are wanted.
- **Disk**: 1.5 TB freed on 17 Oct (validation IC, February checkpoints, BDM PM snapshots); still to decide: the 21 BDM replay density tapes and membership tapes (195 GB on disk), 15 GB of superseded fiducial catalogues.

**Open minors** (grouped in `TRIAGE.md`): parameter-path robustness, $f_R$ product details, provenance JSON, test strength, docs drift. **Questions for Baojiu**: BDM cadence, plane normalisation, CAMB $\omega_c$ pitfall, r2g velocity convention, node 900.

# Next steps

1. Explain the $+0.5\,\%$: the $\mathrm{d}a_0 = 3 \times 10^{-4}$ restart test (half-step term) and a POWMES normalisation check; add per-product timers to `ExtraOutputs` (the V9 split).
2. Settle the three decisions above; rebuild the frozen pilot binaries from `cz`.
3. Production recipe per node: `gen_inputs.py --node N --gravity {lcdm,fr}` $\rightarrow$ `submit.sh` (`ic2pm half` on Baojiu's ICs) $\rightarrow$ `check_run.py`; $\approx 165$ core-hours per $\Lambda$CDM box, $\approx 390$ for f(R); 9 GB ($\Lambda$CDM) or 25 GB (f(R) with the extras) per box.
4. Readers for the new products in `halocat` (planes, modes, subsample, $f_R$), following the `.meta` keys.
5. Reply to Baojiu with the cross-check of the 42 epochs and the open questions.

# Summary

- The DEGRACE companion outputs (schedule, lens planes, $\delta_L$, modes, subsample, $f_R$) are implemented, tested, reviewed and merged into `cz`.
- The $\Lambda$CDM pilot runs in 2.5 h per box with all 42 outputs, the f(R) pilot with all the extras in 6 h; the new products cost about 1 % of a run, the halo finder 23 %, the extras at most 6 %.
- Both pilots agree with the ECOSMOG twin to $+0.5$–$0.8\,\%$ at linear scales at every epoch, and the f(R) run reproduces the f(R) spectrum to $\pm 1\,\%$ up to $k = 0.3$; the small constant offset is common to both gravity models and is the one question still open.
- Three design decisions stand between here and production.